

Personalized Care in CKD: Moving Beyond Traditional Biomarkers

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Abstract

Background: Traditional biomarkers, such as estimated glomerular filtration rate (eGFR) and urine albumin-to-creatinine ratio (uACR), have long been central to chronic kidney disease (CKD) diagnosis and management, leading to a standardized CKD classification system. However, these biomarkers are non-specific and fail to capture the heterogeneity within CKD and the nuances of an individual's disease mechanism, limiting personalized treatment approaches. There is an increasing need for novel biomarkers that reflect the diverse pathophysiological processes underlying CKD progression, enabling more precise risk prediction and treatment strategies. **Summary:** This review examines the limitations of current CKD biomarkers and classification systems, highlighting the need for a precision medicine approach. While traditional markers like eGFR and uACR are foundational, they inadequately capture CKD's complexity. Emerging biomarkers offer insights into specific disease processes, such as inflammation, oxidative stress, fibrosis, and tubular injury, which are crucial for per-

sonalized care. The article discusses the potential benefits of integrating these novel biomarkers into clinical practice, including more accurate risk prediction, tailored treatments, and personalized clinical trial designs, as well as the barriers to their implementation. Furthermore, advancements in multi-omics and high-throughput techniques offer opportunities to identify novel causative proteins with druggable targets, pushing CKD care towards greater precision. **Key Messages:** Current CKD classification systems, based on non-specific biomarkers, fail to capture CKD's heterogeneity. Incorporating biomarkers reflecting diverse pathophysiological mechanisms can enhance risk prediction, customized treatments, and personalized clinical trials. High-throughput multi-omic techniques present a promising path towards precision medicine in nephrology.

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Introduction

The ultimate goal of precision medicine is to tailor treatment to specific disease processes, thereby optimizing patient outcomes [1]. This approach moves

beyond the one-size-fits-all model, recognizing at an individual level the unique combinations of molecular, genetic, and environmental factors determining disease progression and treatment response. Chronic kidney disease (CKD) exemplifies the need for precision medicine, given its complex and heterogeneous nature. Traditional biomarkers, such as estimated glomerular filtration rate (eGFR) and urine albumin-to-creatinine ratio (uACR), have long been the cornerstone of CKD diagnosis and management. However, these markers homogenize a diverse group of distinct conditions within CKD with separate pathophysiologies and progression rates. The standardization of CKD definition has improved clarity and consistency but has inadvertently led to a generic classification system, which categorizes patients with CKD based on these non-specific markers and fails to capture the nuances of individual patient conditions. As a result, there is a critical need for novel biomarkers that can more accurately represent specific aetiologies and mechanisms of CKD progression. By identifying and utilizing novel biomarkers, the field of nephrology could better understand individual mechanisms of CKD progression and move towards tailored risk prediction and treatment strategies, ultimately improving patient outcomes. This review is not intended to be a comprehensive review of all biomarkers in CKD but a proposal to the nephrology community to think more pathophysiologically about CKD, recognize the importance of distinct primary kidney diseases, and start working towards a more personalized medicine approach.

Historical Perspectives and Current Biomarker Definitions of CKD

CKD is inherently tied to biomarkers, with creatinine-based estimations of GFR and albuminuria being central to its definition [2]. The practice of analysing urine dates back to ancient civilizations and the correlation between the physical characteristics of urine and disease can be considered one of the earliest forms of biomarkers [3]. The use of specific biomarkers became integral to the study of kidney disease, with techniques developed to measure serum creatinine in the late 19th and early 20th century later used to estimate GFR [4–7]. However, before the National Kidney Foundation's Kidney Disease Outcomes Quality Initiative (KDOQI) guidelines in 2002, CKD definitions were inconsistent, with over 23 terms used at one American Society of Nephrology meeting alone [8]. This lack of standardization hindered under-

standing of the epidemiology and burden of CKD and impeded clinical trial development, with fewer randomized controlled trials (RCTs) in nephrology than any other internal medical speciality in the 20th century [9] and a lack of effective therapeutics [10].

Through meta-analyses, it became apparent that both uACR and eGFR were independently associated with death and progression of CKD [11]. As a result, they formed the basis of a new unified definition and classification system, first with KDOQI's eGFR-based classification in 2002 [12] and subsequently KDIGO's eGFR/uACR risk-based classification system in 2012 [13] (Fig. 1). This unified consensus allowed subsequent large-scale epidemiological studies, revealing CKD's significant global impact [14, 15]. We now know CKD affects over 840 million people worldwide and is a major risk factor for cardiovascular events and mortality, with projections indicating it will become the 5th leading cause of death globally by 2040 [16]. These current classifications of kidney diseases based on eGFR and uACR emphasize the importance of "common pathway" mechanisms of CKD progression, which are valid therapeutic targets and have been pivotal in providing a rationale for large clinical trials, addressing the shortfall lamented in the early 2000s [9]. In the last 5 years, three new effective medication classes targeting proteinuria reduction have been licensed for CKD, demonstrating significant advancements in therapeutic options [17–19].

Need for Precision in CKD Classification

Despite the benefits of the current classification system, which allows nephrologists to risk stratify patients and use standardized treatments, it overlooks a critical issue. CKD is not a single diagnosis but encompasses multiple heterogeneous conditions, each with unique pathophysiology, mechanisms, and rates of progression [20, 21]. eGFR and uACR are simply generic markers of glomerular filtration and glomerular damage, respectively, revealing little about underlying pathophysiology and failing to capture the diversity of CKD. This conflicts with the principles of precision medicine, where the ultimate goal would be to develop a classification system aligned more closely with pathophysiologic processes. This would then allow treatments tailored to specific underlying disease aetiologies to be developed, selected as appropriate for specific patients and patient outcomes optimized. To exemplify this issue, consider 3 hypothetical patients:

KDIGO: Prognosis of CKD by GFR and albuminuria categories				Persistent albuminuria categories		
				Description and range		
				A1	A2	A3
				Normal to mildly increased <30 mg/g <3 mg/mmol	Moderately increased 30–300 mg/g 3–30 mg/mmol	Severely increased >300 mg/g >30 mg/mmol
GFR categories (mL/min/1.73 m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

Green: low risk (if no other markers of kidney disease, no CKD); Yellow: moderately increased risk; Orange: high risk; Red: very high risk. GFR, glomerular filtration rate.

Stevens PE, Ahmed SB, Carrero JJ, Foster B, Francis A, Hall RK, et al. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int.* 2024 Apr;105(4):S117–314.

Fig. 1. Kidney Disease: Improving Global Outcomes (KDIGO) CKD risk categories [2].

- Patient A: a 42-year-old female with immunoglobulin A (IgA) nephropathy on biopsy, an eGFR of 26 mL/min/1.73 m², uACR of 96 mg/mmol, and no past medical history
- Patient B: a 58-year-old male with type 1 diabetes, an eGFR of 17 mL/min/1.73 m², uACR of 196 mg/mmol, and ischemic heart disease
- Patient C: a 79-year-old lady with an eGFR of 28 mL/min/1.73 m², uACR of 45 mg/mmol, no kidney biopsy, and a background of hypertension, chronic obstructive pulmonary disease, and frailty

Nephrologists would intuitively recognize the heterogeneity in diagnosis and risk among these 3 patients. However, we could also distil each of them down to the same CKD stage, G4A3, and recommend the same medications. The generic nature of these biomarkers is evidenced by the fact that most underlying causes of CKD need to be diagnosed via biopsy. However, this practice comes with risk, and alternative strategies would be preferable [22].

Oncology exemplifies precision medicine, using a plethora of histopathological, protein, and genetic bio-

markers to assist in diagnosis, prognosis, and assessment of treatment response. This is in marked contrast to the approach to caring for individuals with CKD, which has been criticized as a “reductionist” method largely based on generic biomarkers [23].

Limitations of Current Risk Prediction Models Based on Generic Biomarkers

Accurate risk prediction in CKD is crucial for providing optimal care. While only 2% of prevalent CKD patients require kidney replacement therapy (KRT), KRT accounts for over half of renal healthcare expenditure and is associated with significant risks, including reduced quality of life and increased mortality [24]. Precise risk prediction would enable earlier treatment for those at the highest risk of progression and better allocation of resources for transplant workup and pre-dialysis planning.

Several risk prediction models have been proposed for CKD [25]. The 4-variable Kidney Failure Risk Equation (KFRE) is the most well validated among these models,

using age, sex, uACR, and eGFR. The initial study showed high predictive accuracy in a validation cohort with a C-statistic of 0.85 and 0.83 for the 2- and 5-year risk of initiating KRT, respectively [26]. This has been subsequently validated in a large multinational meta-analysis of over 700,000 CKD patients [27]. Its use has been suggested for both referral to nephrology and guiding the timing of vascular access and has been incorporated into international guidance [2].

Though KFRE performs well on a CKD population level, its one-size-fits-all approach fails to recognize or incorporate the diverse mechanisms of progression or primary renal diagnosis, leading to poor performance in several CKD subgroups [27]. In patients with well-preserved renal function, eGFR and uACR predictive accuracy is much lower [28]. KFRE is also less accurate in certain diseases [29]. Compared to other causes, the risk of KRT is underestimated in patients with autosomal dominant polycystic kidney disease (ADPKD), likely because uACR is less representative of the pathophysiology of ADPKD progression [30]. The same is true for people with rare kidney diseases, who make up 5–10% of people with CKD, but they account for over 25% of patients who undergo KRT. The National Rare Disease Registry (RaDaR) showed that the 5-year cumulative incidence of KF was 28 times higher in those with rare kidney diseases compared to those with CKD of any cause, and KFRE consistently underestimated the 2-year risk of KF in this group [31].

The reasons for KFRE's failure to accurately predict risk in these subgroups is multifaceted. Firstly, outside of glomerular damage, the model provides no insight into the biological mechanisms by which CKD progresses in individual patients [32]. This failure is accentuated by the complex mechanism of rare kidney diseases and those dominated by non-glomerular processes such as ADPKD, and to overcome this, more representative biomarkers are needed. Secondly, KFRE does not model for the competing risk of death [33]. CKD in itself increases the risk of death, particularly from cardiovascular causes [34]. However, again, not all causes of CKD carry equal risk. DKD and renovascular kidney disease both carry a much higher risk of death pre-KF than rare causes of kidney disease [31]. If a model does not account for the competing risk of death before the event, it will systematically overestimate the risk of the event in groups with a higher mortality rate [35]. This exemplifies how the unique circumstances of individuals with CKD are overshadowed by generalized models that fail to account for personal and disease-specific variations. In a similar limitation, KFRE artificially constructs the time points of

2- and 5-year risk, yet, as CKD is incurable, for those who are young and with a low risk of death pre-KF, short-term risk prediction is much less meaningful. The prime example is IgA nephropathy. Even with an annual eGFR loss of just 1 mL/min/1.73 m² loss, which many clinicians would consider slow, if a person is diagnosed before their 18th birthday the lifetime risk of KF remains at a staggering 56%, increasing to 100% if GFR loss is 2 mL/min/1.73 m²/year [36]. Lastly, the clinical endpoint of the model is inherently flawed. KFRE predicts risk of starting KRT and not KF, yet the decision to start KRT in a person with KF is subject to significant variability and bias. This is of particular importance when considering risk prediction from a person-centred healthcare perspective as women are less aware of their diagnosis and more likely to choose conservative care than their male counterparts, as are older people [37, 38]. However, despite building sex and age into the model, when there is a decision-based endpoint, assumptions are made about individual choices that may not accurately reflect the true risk of disease progression and reduce the prediction model's precision [35].

Capturing the Heterogeneity of CKD through Representative Biomarkers

To truly embrace precision medicine in CKD, it is imperative to integrate biomarkers that reflect the diverse underlying pathophysiological mechanisms, such as parenchymal cell loss, chronic inflammation, fibrosis, oxidative stress, and tubular injury, all contributing to disease progression [39]. This would enable more personalized and effective prognostic tools and treatments.

The BEST (Biomarkers, Endpoints, and other Tools) resource was developed by the US Food and Drug Administration (FDA) and the National Institutes of Health (NIH) to standardize biomarker terminology and enhance communication among researchers, clinicians, and regulatory bodies [40]. This extends some of the WHO International Programme on Chemical Safety: Biomarkers Criteria in Risk Assessment: Validity and Validation (2001) [41]. It describes in detail the different biomarker roles and outlines key aspects of biomarker development. It is important to clearly define the intended role of the biomarker, which has been broken down into the following categories:

1. Diagnostic
2. Monitoring
3. Prognostic
4. Predictive

5. Pharmacodynamic/response
6. Safety
7. Susceptibility/risk

An important distinction to make is the difference between a prognostic biomarker, which provides information about the likely course or outcome of a disease, and a predictive biomarker, which indicates the likely response to a specific treatment. Each individual biomarker should also fulfil the following criteria:

1. Biological plausibility: the biomarker should be clearly and understandably connected to the biological mechanisms of the disease or condition being studied, providing a rationale for its use and paving the way for companion diagnostics.
2. Measurability: it should be possible to measure the biomarker accurately and reproducibly using standardized methods and available technology.
3. Valid measures of performance assessment: these are dependent on the type of biomarker and intended to be used clinically such as in diagnosing, prognosticating, predicting disease, and/or monitoring treatment response. Analytical specificity and sensitivity, precision, and accuracy are key analytical metrics of assay performance of clinical relevance, which then need to be complemented by clinical sensitivity, specificity, and negative or positive predictive values.
4. Validation: this includes both analytical validation of the biomarker measurement and clinical validation in terms of ensuring that it reliably predicts the clinical outcome assessments across different populations and settings.

Hence, a biomarker should have a clear and understood connection to the biological mechanisms underlying a disease. Within CKD, albuminuria serves as a primary marker of glomerular damage [42, 43]. Interventions with renin-angiotensin system blockade, sodium-glucose cotransporter-2 inhibitors (SGLT2i), and mineralocorticoid receptor antagonists reduce intraglomerular pressure, proteinuria, and glomerulosclerosis, consequently slowing CKD progression [44]. Based on available evidence and biologically plausible mechanisms, albuminuria has been recommended as a surrogate endpoint for diabetic kidney disease (DKD) and IgA nephropathy [45]. However, for conditions not characterized by albuminuria, such as polycystic kidney disease or tubulointerstitial diseases, more biologically plausible biomarkers, such as total kidney volume or tubular injury markers, have been suggested.

There has been increasing interest in the study of novel blood and urine biomarkers in CKD over the past 20 years. Dedicated longitudinal CKD cohorts, such as

the Canadian Study of Prediction of Death, Dialysis and Interim Cardiovascular Events (CanPREDDICT), the Chronic Renal Insufficiency Cohort (CRIC) Study, and the National Unified Renal Translational Research Enterprise (NURTuRE), have been established to facilitate this [46–48]. Evidence now supports the role of various biomarkers in reflecting the diverse mechanisms driving CKD progression, including nephron loss, inflammation, myofibroblast activation, extracellular matrix (ECM) deposition, and lipotoxicity-induced oxidative stress [49]. It is important to “think pathophysiologically,” and the following section (as summarized in Fig. 2) highlights measurable biomarkers associated with the main pathophysiological processes and explores novel therapeutic options targeting these pathways. This summary does not aim to provide a comprehensive review of all CKD biomarkers but rather to offer an overview of some of the most promising emerging biomarkers. For those seeking more in-depth information, there are several comprehensive reviews on CKD biomarkers [50–52].

Inflammation

Inflammation is key in CKD progression, driven by pro-inflammatory cytokines and immune cells. Macrophages are central to this process, producing reactive oxygen species (ROS) and cytokines such as tumour necrosis factor- α (TNF- α) and interleukin-1 beta (IL-1 β), exacerbating inflammation and kidney damage. Sustained inflammation leads to fibrosis and excessive ECM deposition, covered in more detail below [49]. Key signalling pathways mediate these inflammatory responses, such as nuclear factor-kappa B (NF- κ B) and Janus kinase-signal transducer and activator of transcription (JAK-STAT). Interleukin-6 (IL-6) and monocyte chemoattractant protein-1 (MCP-1) are significant contributors, with TNF- α activating NF- κ B, thereby increasing pro-inflammatory gene transcription. Mediators such as soluble TNF receptor 1 (sTNFR1), which binds to TNF- α , can mitigate pro-inflammatory effects by preventing TNF- α from interacting with its cell surface receptors. However, depending on the levels and duration of upregulated TNF- α , soluble TNFR1 can also stabilize TNF- α , potentially prolonging its activity [49].

sTNFR1 has been extensively examined as a prognostic biomarker in CKD and can be measured via commercially available ELISA or as part of a Meso Scale Discovery (MSD) multiplex immunoassay platform and has been shown to both predict CKD progression and mortality in those with CKD [53, 54]. IL-6 can similarly amplify inflammatory and fibrotic processes, playing a direct role in the development of cardiovascular disease [55]. It is

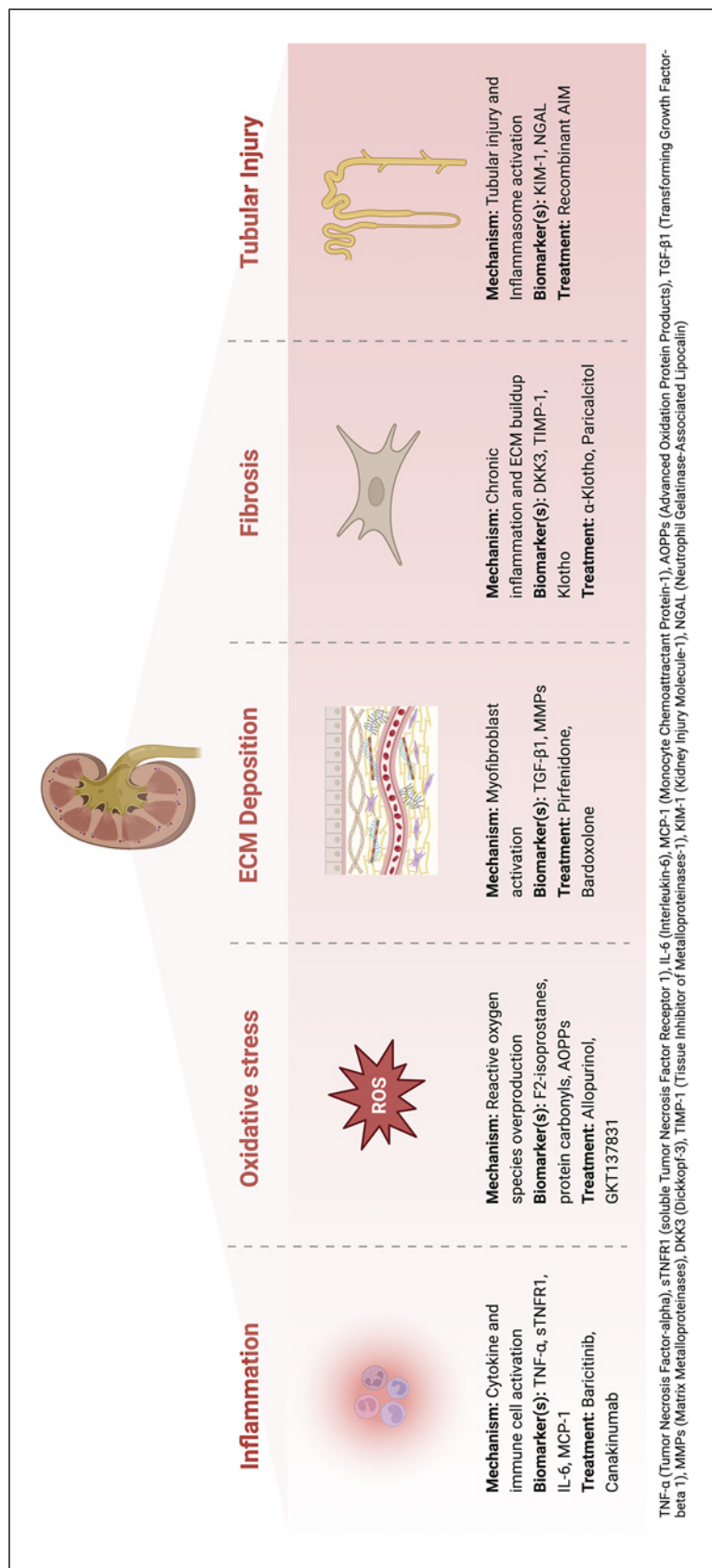


Fig. 2. Mechanisms of CKD progression, biomarkers, and potential therapeutics.

reliably measured in serum, and strong evidence supports its use as a prognostic biomarker for cardiovascular events in CKD [56]. MCP-1 attracts monocytes to the site of inflammation, further amplifying the inflammatory response and showing potential as a diagnostic and prognostic biomarker in DKD strongly associated with albuminuria, eGFR decline, and risk of CKD progression. Plasma MCP-1 also predicts adverse renal and cardiovascular outcomes, particularly in advanced stages of DKD [49, 57].

Importantly, these biomarkers not only represent key pathophysiological pathways in CKD and represent diagnostic and prognostic markers, but they also have actionable targets. In a study of those with DKD, baricitinib, a JAK-STAT pathway inhibitor, significantly decreased plasma sTNFR1 and urinary MCP-1 levels in a dose-dependent manner supporting their potential role as response biomarkers [58]. Several novel CKD treatments specifically targeting and inhibiting inflammation have shown promising results in animal models. Biomarkers can thus serve in prediction and assessing the extent of disease, and they can also highlight novel targets for drug repurposing [49]. canakinumab, an IL-1 β inhibitor, whose role in inflammation and the pathogenesis of cardiovascular disease is well established, was assessed for its effects on cardiovascular outcomes in patients with CKD. In this study, inhibition of IL-1 β with canakinumab effectively reduced cardiovascular events in high-risk CKD patients. Importantly, patients with higher baseline levels of inflammation, as defined by levels of hsCRP and IL-6, had a greater cardiovascular risk, highlighting their benefit as prognostic biomarkers, whilst those achieving significant reductions in hsCRP (to levels <2 mg/L) experienced the most pronounced benefits (response biomarker) when receiving canakinumab [59].

Oxidative Stress

Oxidative stress plays a crucial role in the progression of CKD through the overproduction of ROS such as superoxide anion, hydrogen peroxide, and hydroxyl radicals, primarily generated by NADPH oxidase (NOX) and xanthine oxidase. Key signalling pathways involve the upregulation of Nox4 in the kidney, leading to excessive ROS accumulation and activation of downstream pathways like ERK, mTORC, and NF- κ B, contributing to chronic inflammation and renal fibrosis [49].

The RENIS study explored the predictive value of oxidative stress biomarkers, specifically urinary 8-oxodG and 8-oxoGuo, as potential early diagnostic indicators of kidney dysfunction. Higher urinary levels of 8-oxoGuo

were independently associated with an increased risk of developing low-grade albuminuria, highlighting its potential as a sensitive diagnostic biomarker. However, the study used liquid chromatography-tandem mass spectrometry, a method that may not be easily implementable in routine clinical practice [60]. F2-isoprostanes are also reliable biomarkers for assessing oxidative stress and lipid peroxidation in vivo and they have efficacy as diagnostic markers in CKD. However, their measurement also relies on mass spectrometry, which is precise but costly and time-consuming, limiting its use in routine clinical practice [61, 62]. Other measurable biomarkers of oxidative stress are advanced oxidation protein products [63]. In cross-sectional analysis, advanced oxidation protein products levels were higher in advanced CKD. However, commercial assays are not available.

Therapeutics aimed at reducing oxidative stress have been explored. While initial studies with xanthine oxidase inhibitors like allopurinol showed some efficacy in slowing kidney function decline [6], larger RCTs such as PERL and FEATHER, which examined allopurinol and febuxostat, respectively, found no significant differences in the rate of eGFR decline despite sustained reductions in serum uric acid levels. Although uric acid is known to act as a scavenger of free radicals and ROS, and could serve as a response biomarker for oxidative stress modulation, the lack of clinical benefit observed suggests that lowering serum uric acid alone may not adequately capture the broader oxidative stress and inflammatory processes driving CKD progression [64, 65]. Novel therapeutics, including NADPH oxidase inhibitors like GKT137831, have shown promise in reducing albuminuria and nephropathy in diabetic models [66].

Extracellular Matrix Deposition

The deposition of ECM is a critical driver of CKD development and a marker of progression. ECM components such as collagen and fibronectin accumulate excessively when myofibroblasts are activated through pathways like TGF- β , Wnt/ β -catenin, and Notch [49]. This ECM build-up leads to structural changes and impairs kidney function. Key biomarkers for ECM deposition include TGF- β and matrix metalloproteinases such as MMP-2, MMP-7, and MMP-9, which are involved in ECM turnover and have been shown to correlate with worsening eGFR, proteinuria, and renal fibrosis [67, 68]. MMP-7 is an enzyme-secreted endopeptidase that relies on zinc and calcium to degrade various substrates within the ECM and basement membrane [69]. It is a downstream target gene of the

Wnt/ β -catenin signalling pathway and plays a crucial role in kidney fibrosis by promoting the breakdown of ECM components, activation of pro-fibrotic molecules (including TGF- β), leading to tissue remodelling and fibrotic scarring [70].

TGF- β 1, measured by ELISA, demonstrated superior predictive accuracy compared to eGFR and uACR for the progression of DKD, highlighting its potential utility as a prognostic biomarker [71]. Therapies targeting ECM deposition include pirfenidone and TGF- β signalling inhibitors. Plasma and urinary TGF- β , measured using the MSD platform and ELISA, respectively, were both associated with lower eGFR, suggesting a pathophysiological link. However, despite pirfenidone demonstrating efficacy in slowing CKD progression, there were no significant changes in TGF- β levels in either plasma or urine following treatment [49, 72]. An ongoing phase 2 TOP-CKD trial (NCT04258397), aiming to recruit 200 people with an eGFR ≥ 20 mL/min/1.73 m² at risk of progression, will assess the effectiveness of pirfenidone to slow CKD and also measure urinary fibrotic biomarkers to assess their response to treatment and implications for progression.

More novel therapeutics targeting TGF- β , such as anti-TGF- β monoclonal antibodies, are in the trial phase of development [73]. Other novel inhibitors, such as bardoxolone, which activates Nrf2 and increases Smad7 levels to inhibit TGF- β signalling, show promise in reducing ECM deposition and renal fibrosis when assessed by tissue analysis [74]. However in the BEACON trial, bardoxolone did not reduce the risk of kidney failure (KF) or death and there was a higher rate of cardiovascular events [75].

Fibrosis

Fibrosis represents an irreversible process of tissue damage and can be considered a final “common pathway” resulting from mechanisms such as inflammation and oxidative stress, which significantly limit clinicians’ available therapeutic options [76]. The extent of tubulointerstitial fibrosis in renal biopsies is a good predictor of kidney survival in patients with CKD [77]. However, currently, the only validated tool to assess the severity of tissue damage is the kidney biopsy [78]; yet this is an invasive procedure, not without risk, and serial biopsies are justified only in very specific circumstances [52].

The pathogenesis of fibrosis is closely related to that of ECM deposition. Though renal fibrosis may correlate with albuminuria and eGFR, a number of specific biomarkers of renal fibrosis have been explored, including tissue inhibitor of metalloproteinases 1 (TIMP-1), which

has been implicated in age-related renal fibrosis [79]. Also, urinary Dickkopf-3 (DKK3), a stress-induced glycoprotein released by tubular epithelial cells, induces tubulointerstitial fibrosis through the Wnt/ β -catenin pathway. It can be measured via commercially available ELISA and promisingly, as it correlates with biopsy-proven tubulointerstitial fibrosis, it could be employed as a diagnostic biomarker providing non-invasive information as to the extent of renal fibrosis as well as a prognostic biomarker predicting CKD progression [80, 81].

Another biomarker of renal fibrosis is klotho, an anti-ageing protein highly expressed in kidney tubular epithelia, which functions as a co-receptor for FGF23 in phosphate homeostasis. Its soluble form is involved in various protective mechanisms against CKD progression by inhibiting TGF- β , Wnt, and FGF23 signalling pathways [82]. Klotho deficiency has been postulated as a driving force of CKD progression [83]. It has been studied extensively as a prognostic biomarker in CKD, with lower levels linked to increased risks of CKD progression, KF, and higher cardiovascular and all-cause mortality [84].

Given that low klotho expression is associated with worse outcomes, exogenous α -klotho has itself been tested as a novel treatment targeting the Wnt/ β -catenin pathway as it inhibits Wnt signalling. This and paricalcitol, which suppresses β -catenin-mediated transcription, have shown effectiveness in reducing fibrosis [82]. Furthermore, Dickkopf (DKK) family proteins, natural antagonists of Wnt/ β -catenin signalling, offer antifibrotic potential by binding to LRP6 and blocking Wnt ligand interactions, with markers of the WNT- β -catenin pathway acting as potential response biomarkers [85].

Tubular Injury

Tubular injury is a common biopsy finding, and the degree of tubular injury on biopsy correlates with CKD progression [77]. The kidney’s tubules make up 90% of its functional units and account for the majority of its energy usage. However, the current model of CKD remains “glomerulocentric,” with biomarkers of tubular injury not commonly used in clinical practice and tubular damage being invisible to clinicians without a biopsy. Mechanisms of tubular injury in CKD are often related to the aetiology, and tubular injury itself can lead to the activation of inflammasomes and the release of cytokines, contributing to fibrosis and CKD progression [49]. Tubular markers can provide prognostic information on CKD progression, CVD, and mortality, even after accounting for traditional CKD risk factors. These markers

can reflect processes of direct tissue injury, repair, inflammation, tubule cell protein overload, and fibrotic activity within the kidney [86]. Examples are kidney injury molecule-1 (KIM-1) and neutrophil gelatinase-associated lipocalin (NGAL), markers of proximal and distal tubular injury, which have been extensively studied in CKD, and both have commercially available assays [53]. KIM-1 can modulate and strongly predict disease progression, particularly in cystic conditions [87, 88]. NGAL has been examined as a diagnostic biomarker in CKD, highlighting those at risk of early kidney function decline, a sensitive marker of tubular injury in IgA nephropathy and systemic lupus erythematosus, as well as correlating with cyst size in ADPKD. It also operates as a prognostic biomarker, which is considered an independent predictor of adverse renal outcomes and death [89]. Yet, despite providing insights into nephron injury outside the glomerulus, they have yet to be incorporated into routine clinical practice.

This may be because markers of tubular dysfunction, like many biomarkers in CKD, are usually assessed individually to determine if they can predict the risk of KF [90], rather than considering the additional information these markers might provide regarding other disease mechanisms or responses to treatment. However, if we are to focus solely on whether a tubular biomarker can improve the prediction of CKD progression, studies have shown that using a combination of biomarkers can improve discriminatory metrics [91].

Biomarker Panels

Given the complex and multifaceted nature of CKD pathophysiology, a single biomarker is unlikely to capture the diverse underlying processes necessary for accurate disease assessment. While clinicians may opt for a tailored panel of biomarkers based on clinical judgement, advances in highly multiplexed profiling techniques offer alternative broader views. This is exemplified by CKD273, a support vector machine classifier based on 273 urinary peptides detected using capillary electrophoresis coupled with mass spectrometry [92, 93]. This panel reflects key pathophysiological processes of CKD, encompassing a diverse array of peptides, including collagen fragments, markers of tubular injury, and inflammatory proteins.

CKD273 has been extensively studied in both diagnostic and prognostic contexts in CKD. In the PRIORITY and DIRECT-Protect 2 studies, a high risk CKD273 score classification was associated with increased risk of progression to microalbuminuria in individuals with type 2 diabetes, demonstrating its utility as an early diagnostic marker for DKD [94, 95]. Furthermore, CKD273 has

shown prognostic value, predicting CKD progression [96] and outperforming albuminuria in predicting eGFR decline in individuals with early-stage DKD (eGFR >70 mL/min/1.73 m²) [97]. This is particularly significant as albuminuria and eGFR are markers of late-stage disease and general kidney damage. In contrast, biomarker panels like CKD273 may offer insights into the early pathological changes occurring within the kidney.

6.

Application of Novel Biomarkers in Personalized CKD Care

Emerging biomarkers that capture the diverse mechanisms of CKD progression and are linked to therapeutic benefits are now available, and integrating these biomarkers into clinical practice can help capture the heterogeneity within CKD. This approach mirrors the success of precision medicine in oncology, emphasizing the importance of personalized treatment strategies for optimal patient outcomes. We discuss this further below in the context of prognosis/risk prediction, treatment, and clinical trials, as shown in Figure 3.

Ideal Risk Prediction Model

There is a fundamental tension between population-based and individualized treatment approaches for CKD. While a universally applicable, parsimonious model is highly appealing, such models are most practical in primary care settings, where the specific cause of CKD is often unknown, and broad treatments targeting the common pathways of CKD progression can be applied to many people, according to established guidelines. The KFRE performs well when considering CKD as a whole but, as described previously, is limited individually. This is underscored by the high negative predictive value but low positive predictive value of the KFRE, indicating its suitability as a tool to guide referral rather than a nuanced treatment guide [98]. More tailored models are essential in the specialist office, where the cause of CKD is known and the patient's unique characteristics are better understood. The ideal model of personalized risk prediction should incorporate multiple data points in addition to serum creatinine and urinary albumin (Fig. 3). When considering a person with CKD, a plethora of additional pathological biomarkers (mentioned previously) could be measured. These biomarker profiles would provide the clinician with an understanding of the "phenotype of CKD," for instance, inflammatory or fibrotic, as well as markers of tubular dysfunction, helping to guide a more nuanced treatment approach and holistic understanding

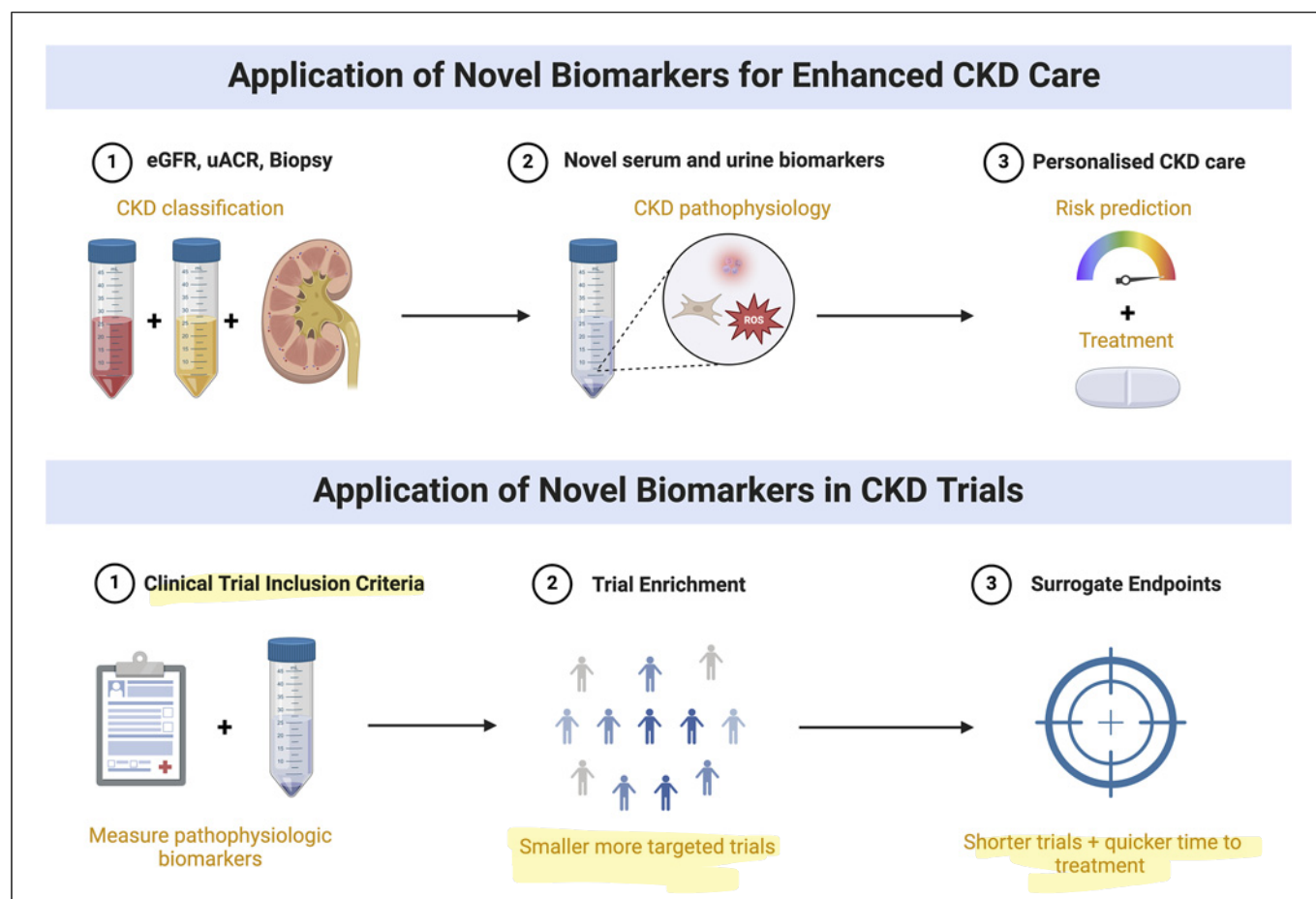


Fig. 3. Proposed application of novel biomarkers for enhanced CKD care and clinical trials.

of the kidney's function. The primary renal diagnosis should also be incorporated into the risk prediction model, either obtained by clinical history, kidney biopsy, or specific diagnostic biomarkers already developed in glomerulonephritis [99–101]. Lastly, the use of multi-omic techniques, discussed later, such as genomics, pathomics, and metabolomics, could provide an even clearer window into the individual, capturing the true uniqueness and allowing personalized prediction.

Despite the wide array of biomarkers representing multiple mechanisms in CKD, advancing risk prediction beyond traditional risk factors remains challenging. Methodological, cultural, and cost-related issues (discussed in detail in the following section) have hindered progress in biomarker research for CKD. However, it is likely that single biomarkers are insufficient for accurately predicting outcomes, whereas panels of biomarkers representing multiple pathways may offer greater predictive power [102]. Furthermore, the advent of multi-omic techniques – such as

those demonstrated by CKD273 – holds promise for further enhancing our ability to predict CKD progression.

It should also be considered that the current risk prediction models in CKD mainly rely on prognostic biomarkers to predict the likelihood of CKD progression and KF. However, there is potential for more personalized risk prediction to be utilized in tailoring and predicting responses to treatments, similar to the approach commonly used in oncology and personalized clinical trial enrolment, both of which we will discuss below.

These biomarkers would not only guide treatment decisions but also monitor treatment response. While this may sound futuristic, as discussed, it is already a reality in oncology [103].

Biomarkers to Monitor Response to Treatment

Ideally, biomarker levels would not only provide information on the risk of an event but also be modifiable with treatment, with changes in biomarker levels correlating with

changes in the risk of the event as a response to the treatment [32]. Within the current CKD risk model (exemplified by KFRE), only albuminuria is modifiable. However, as a consequence of the medications used to lower albuminuria, doing so by reducing glomerular pressure, the ability to predict risk with changing biomarker levels can be flawed. Take the case of a 48-year-old male with diabetes, an eGFR of 41, and uACR of 180 mg/mmol on no disease-altering medications. His KFRE 5-year risk is 11.9%. Over a period of 6 months, he is treated with maximum tolerated dose RAASi and SGLT2i, and he experiences an 8 mL/min drop in eGFR to 33 (<20% and within “acceptable range”), and his uACR drops from 180 to 28 mg/mmol. This would be considered successful. However, his KFRE 5-year risk prediction would increase marginally from 11.9% to 12.5%. In reality, evidence from trials indicates that his risk of CKD progression and cardiovascular mortality would indeed decrease. Yet our current models are not yet nuanced enough to deal with such modifiable risk.

The ESCAPE trial, an RCT looking at the impact of intensified blood pressure control on the progression of CKD in children, measured the novel urinary biomarker DKK3 at regular intervals [104]. The study found that higher levels of urinary DKK3 were linked to a more significant decline in eGFR over 6 months, independent of other clinical variables. Importantly, those with elevated urinary DKK3 levels also experienced greater benefits from RAASi, highlighting its utility as a “response biomarker.” Furthermore, treatment with RAASi resulted in reductions in urinary DKK3 levels, indicating that this biomarker not only predicts disease progression and response but can also be modified with specific therapeutic interventions and used to monitor treatment.

The biomarker DKK3 provides important information about kidney disease processes that occur outside the glomerulus. It acts as a WNT- β -catenin modulator, promoting kidney fibrosis and serving as a new marker for CKD progression. DKK3 is released into the urine during acute or prolonged stress on the renal tubules. Therefore, the response of this biomarker to RAASi can provide insights beyond haemodynamic changes in the glomerulus, acting as a potential marker of kidney damage. Importantly, several inhibitors targeting the WNT- β -catenin signalling pathway are in development [85].

The CKD273 urinary peptide profiles have also been shown to change in response to treatment. Although there were small numbers of patients, following the initiation of irbesartan, CKD273 profiles showed a shift from CKD towards the “healthy” classifier. These alter-

ations may reflect underlying pathophysiological processes, as indicated by the greatest reductions seen in levels of collagen fragments [105]. CKD273 has also been investigated as a response biomarker in DKD and mineralocorticoid receptor antagonist use. In a post hoc analysis of an RCT involving patients with diabetes, albuminuria, and resistant hypertension, those with the highest CKD273 scores showed the greatest benefit from spironolactone, particularly in reducing proteinuria [106]. The PRIORITY study further explored whether high risk CKD273 could predict the response to spironolactone in diabetic patients with normoalbuminuria. While higher CKD273 scores were associated with progression to microalbuminuria, they did not identify patients who would respond to spironolactone, likely because spironolactone did not significantly reduce the progression to microalbuminuria overall [94]. By utilizing bespoke biomarkers that reflect individualized disease progression mechanisms, CKD care can significantly enhance the precision of risk prediction, personalize targeted treatments, and improve clinical trial outcomes. This shift from a “one-size-fits-all” approach to a more personalized methodology promises to transform nephrology into a field where therapies are tailored to each patient’s unique needs.

Biomarkers in Clinical Trials

Using “clinically meaningful” endpoints in clinical trials is essential to ensure that developed medications provide meaningful benefits to patients. In CKD, such endpoints include KF (necessitating dialysis or symptom-inducing levels of kidney dysfunction) and mortality. However, the slow progression of KF means that clinical trials using these endpoints must either be prolonged or limited to patients with advanced CKD, whose pathophysiological processes may differ from those in early-stage disease. These constraints have made nephrology trials historically less attractive to pharmaceutical companies, and for a long period, there were very few large clinical trials and limited therapeutic options available to CKD patients [9]. Applying a biomarker as a surrogate endpoint in a clinical trial, such as albuminuria, serum creatinine doubling, or eGFR slope, allows earlier detection of treatment effectiveness. However, to ensure clinical relevance, these surrogates must reliably predict clinical outcomes and be supported by biological plausibility and evidence from epidemiological studies [107], with a causal link established [108]. The use of eGFR slope and albuminuria as surrogate endpoints was approved for DKD and IgA nephropathy [45] and has completely revolutionized the landscape for the latter.

IgA nephropathy often affects young individuals, and until recently, treatment options were extremely limited, leading to an almost inevitable progression to KF in those with severe disease [36]. The extended time to KF made the prospect of clinical trials lengthy and costly, deterring pharmaceutical companies from investing in new treatments. However, following the approval of eGFR slope and albuminuria as surrogate endpoints in 2018 we now have two new effective medications: sparsentan (PROTECT [109]) and targeted-release budesonide (NeflgArd [110]), which effectively slow the progression of IgA nephropathy. While albuminuria remains a generic marker of glomerular damage in IgA nephropathy, measuring pathogenic galactose-deficient IgA1 (Gd-IgA1) as a biomarker [111], as done in the NeflgArd trial, shows promise. The trial demonstrated dose- and time-dependent reductions with treatments [112], highlighting the potential for more specific biomarkers to assess disease activity and treatment response. The next step is to establish a link between changes in these pathogenic biomarkers and clinical outcomes in trials, similar to those for albuminuria.

Biomarkers have advanced even further in the context of IgA nephropathy. Through multiple studies, we understand the importance of B-cell activating factor, a proliferation-inducing ligand [113], and complement pathways [114]. There are now over 20 agents targeting B-cell production of Gd-IgA1 and complement activation [113, 115], with related biomarkers being measured in these trials [116]. This comprehensive approach to biomarker research in IgA nephropathy exemplifies how bespoke use of biomarkers in clinical trials can revolutionize treatment paradigms. It serves as a blueprint for CKD treatments and clinical trial development, demonstrating the potential to personalize and optimize patient care.

Not only can the measurement of biomarkers that relate to the direct pathogenic mechanism enable assessment of treatment response, but it can also lead to targeted and personalized clinical trials. Biomarkers can enrich trial populations by selecting individuals who are more likely to respond to treatment, known as enrichment or stratified medicine trials [117]. Prognostic enrichment involves enrolling participants with biomarkers associated with higher event probabilities, thereby reducing participant numbers and trial duration and cost. Predictive enrichment selects individuals more likely to respond to treatment, enhancing trial efficiency and ensuring the right person receives the right drug. For instance, the SONAR study of atrasentan in DKD used an enrichment strategy by enrolling patients with specific

uACR levels and identifying responders based on uACR reduction and tolerability during a run-in phase [118].

With the development of biomarkers that are specific to the action of the drug, such as Gd-IgA1 with budesonide, B-cell depleting agents, or urinary complement levels in narsoplimab (an MASP2 inhibitor) [119] and iptacopan (a factor B inhibitor) [120], the future of patient-centred trials in nephrology is bright with the potential for clinicians to measure an array of pathogenic biomarkers and choose targeted therapies accordingly. Even 10 years ago, these developments regarding IgA nephropathy could not have been imagined, but exciting progress has been achieved and we should seek to follow a similar path for other forms of CKD.

Challenges and Future Directions

Methodological and Cultural Challenges

Despite the significant research in this area, using personalized pathophysiological biomarkers and risk prediction models in CKD remains a considerable distance away from routine clinical practice, with a number of potential stumbling blocks from biomarker discovery to implementation. During the biomarker discovery phase, research groups identify candidate biomarkers using many omics-based molecular techniques. However, once a biomarker is identified, a representative CKD cohort is needed to evaluate its association with clinically meaningful outcomes and to enable analytical and clinical validation. It is here that large, well-phenotyped CKD biorepositories, such as NURTURE, CRIC, and CanPREDDICT, are critical [46–48]. These biorepositories provide high-quality samples and associated clinical data and long-term follow-up, which are essential for biomarker research and reduce waiting periods for validation, thus accelerating the research process.

To ensure biomarker studies are effective, standardization and validation between them are essential. Despite considerable effort and investment, substantial heterogeneity in study designs and analytical techniques limits comparability, particularly in the prognostic performances [121]. The BEST toolkit provides a comprehensive framework for addressing these issues by offering standardized definitions and guidelines for biomarker validation [40]. It emphasizes the importance of rigorous scientific evidence to demonstrate that a biomarker reliably predicts clinical outcomes, including standardizing assay techniques. For example, some biomarkers like CD40L are extremely vulnerable to pre-analytical factors,

leading to inconsistent results even within the same cohort if not handled correctly [122, 123].

Even after extensive validation, collaboration and “buy-in” from industry partners are needed to generate commercially available assays. Once this has been achieved, multicentre and real-world validation can take place.

Model development must also be openly reported. Meta-analyses of risk prediction models have noted poor reporting of model development aspects, including which predictors were examined and the model-building strategy [124, 125]. The “Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis” (TRIPOD) statement addresses this, providing a checklist of 22 essential items for good reporting of studies developing or validating multivariable prediction models. Future studies should adhere to the TRIPOD statement to enhance published prediction models’ transparency, completeness, and reproducibility.

When publishing on model performance, the area under the receiver operating characteristic curve is often quoted as the metric of model discriminatory performance. However, the practice of area under the receiver operating characteristic curve “hacking,” where results are manipulated to surpass specific thresholds, undermines the reliability of these prediction models [126]. To avoid such a practice, the model should be tested in an independent validation cohort to ensure its robustness and generalizability. In addition, we propose that additional clinically relevant performance metrics, including positive and negative predictive value, should be evaluated in relevant populations [98].

Finally, in parallel with identifying and validating potential new biomarkers, an additional obstacle to navigate is a cultural one. The nephrology community needs to embrace the move away from solely using the generic classification of CKD in specialist nephrology clinics. Embracing personalized CKD care means recognizing the limitations of our current models and being willing to adopt new, more precise biomarkers that reflect individual disease mechanisms.

Integrating Novel Biomarkers into Clinical Practice: Cost and Practical Considerations

Integrating novel biomarkers into clinical practice, particularly for CKD, presents significant challenges in terms of cost and practicality. With the multiple biomarkers needed to capture CKD complexity effectively and either a single multifaceted prediction model or multiple parsimonious models that can be applied to individual subsets of CKD, the process will be inherently

expensive. Cost implications extend beyond the initial research and development phases to include the implementation of (multiple) biomarker assays in routine clinical settings. Health economic evaluations will be essential to assess how cost-effective these biomarker panels will be. This includes considering the long-term benefits of more precise risk prediction and personalized treatment plans, which could potentially reduce overall healthcare expenses by preventing disease progression and reducing demand for costly KRT. Therefore, while integrating novel biomarkers holds promise for advancing CKD care, it necessitates thorough economic evaluation studies and a strategic approach to managing the associated costs effectively [127].

Future Trends: Multi-Omics, High Throughput Techniques, and Novel Drug Discovery

In terms of increasing our understanding of the biology of CKD and identifying novel biomarkers and potential pathways for therapeutic targeting, unsupervised omics profiling approaches offer opportunities to generate new knowledge which is not based on pre-conceived targeted approaches and existing knowledge. If an observational study suggests a link between a biomarker and KF or CKD progression, we cannot conclude with certainty whether the biomarker directly contributes to the decline in kidney function or if its elevation is a consequence of another underlying process. The advent of high-throughput genotyping platforms has revolutionized genome-wide association studies, identifying over 800 genetic loci associated with GFR [128]. Fine mapping, using larger samples, denser genotyping, and multi-ancestry approaches, has refined these genome-wide association study signals and facilitated the identification of exact causal genes [129]. Through proteome-wide mendelian randomization (MR), truly causal protein biomarkers can be pinpointed via multi-omic strategies, thereby advancing our understanding of disease mechanisms and aiding in novel drug discovery [130]. Proteome-wide MR works by identifying protein quantitative trait loci (pQTL), genetic variants associated with an increased concentration of a protein/biomarker. The correlation between the impact of each pQTL on the altered biomarker concentration is then assessed, and its effect on an outcome, such as KF or CKD progression, is evaluated. As a person is randomly assigned pQTL at birth, these variants act as “natural RCTs” whereby altered biomarker concentrations can be assessed, independent of other confounding factors that might affect the outcome. Given large enough sample sizes, causal links can thus be identified [131].

Liquid chromatography-mass spectrometry-based profiling has already been used in proteomic studies in nephrology to identify key biomarkers in kidney diseases, such as anti-PLA2R antibodies in membranous nephropathy, leading to more precise disease classifications and new diagnostic tools [101]. Alternative proteomic profiling platforms such as SomaScan and Olink have higher throughput and allow several thousand proteins to be analysed simultaneously, in large-scale studies such as those in 35,559 Icelanders and 54,306 UK Biobank participants [132]. Such platforms are now being employed in systematic large-scale CKD studies [99, 133]. Emerging findings from proteomic studies include the identification of kidney risk inflammatory signature proteins contributing to the inflammatory process underlying progression to KF [134] and the identification of causal biomarkers, which improve risk prediction [32, 129].

Excitingly, proteomic analysis in these studies has revealed biological pathways and proteins with druggable targets. Dubin and colleagues undertook a proteome-wide MR study of the CRIC study, validated in the Atherosclerosis Risk in Communities Study (ARIC), where they discovered 65 causal proteins, 5 of which were previously undiscovered. Via cross-referencing the Druggable Target Database, 16 of the known and 2 of the unknown proteins were found to have known druggable targets. Two new proteins, IL-1 sRII and MMP-7, had therapeutics GSK-1827771 and Prinomastat already in development that could be potentially repurposed for trials in CKD [32].

As discussed earlier, biomarkers can be used to personalize clinical trial designs. Via the use of these high throughput techniques, this has already happened in the field of oncology, resulting in trial designs shifting from being tumour-type-centred to gene-directed “histology-agnostic” trials [135]. These trials use innovative adaptive designs and personalized combination treatment strategies, moving away from a one-size-fits-all approach to tailored combinations of therapies based on individual biomarker profiles. This is possible due to the deep phenotyping that occurs within oncology, where genetic, biological, and immune markers are profiled for each patient. Innovative trials such as the Lung-MAP trial, an “umbrella trial” in which multiple treatments are tested in different genomic/biomarker subsets for a single histology, enhance the efficiency of drug development [136]. The NCI-MATCH trial focuses on targeting specific molecular alterations across various tumour types, offering any number of targeted treatments to a patient, de-

pending on what is most promising [137]. Even N-of-1 trials customize treatment plans for individual patients by using a deep understanding of their tumour biology and comparing it to historical data as a control [135]. These trials emphasize patient-centred approaches, where therapies are directed at each patient’s unique genetic and molecular characteristics in a bespoke manner, leading to more effective and personalized treatment strategies.

With the growing availability of large CKD omics datasets and more refined high-throughput platforms, the ability to unmask the molecular heterogeneity within CKD, identify causative proteins with druggable targets, and personalize CKD trial design will only increase.

Conclusion

The standardization of CKD definitions has improved clarity and consistency but has inadvertently led to a generic classification system. This system, based on non-specific biomarkers, fails to capture the nuances of individual patient conditions, overlooking the significant heterogeneity within CKD. Recognizing this diversity is key to personalized medical care. Broad definitions and risk prediction tools are effective in large, undifferentiated cohorts, such as in primary care. However, models that acknowledge and accommodate patient variability are essential for individualized CKD care.

To achieve this, we must adopt a pathophysiological perspective and recognize CKD heterogeneity. We now have a range of established serum and urine biomarkers representing the varied pathophysiological mechanisms of CKD progression, many with available druggable targets. Incorporating these biomarkers into the assessment and management of individuals with CKD provides the opportunity to better understand individual pathophysiology, enabling more accurate risk prediction, customized treatments, and more personalized clinical trial enrolment. This will involve greater standardization in biomarker studies and the utilization of large, well-phenotyped CKD biorepositories.

Finally, by leveraging high-throughput multi-omic techniques and cross-referencing causative proteins with known druggable targets, nephrology has the opportunity to advance rapidly. Embracing these advancements holds the potential to move CKD care into an era of precision medicine, ultimately improving outcomes for individuals with CKD.

Conflict of Interest Statement

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Thomas McDonnell devised the concept and wrote the manuscript. Rosamonde E. Banks, Maarten W. Taal, Nicolas Vuilleumier, and Philip A. Kalra contributed to the concept and edited the manuscript. All co-authors reviewed and revised the manuscript, and the final version was approved for publication by all co-authors.

References

- 1 Precision medicine in nephrology. *Nat Rev Nephrol* 2020;16(11):615.
- 2 Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group, Stevens PE, Ahmed SB, Carrero JJ, Foster B, Francis A, Hall RK, et al. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int.* 2024;105(4S):S117–314. <https://doi.org/10.1016/j.kint.2023.10.018>
- 3 Connor H. Medieval uroscopy and its representation on misericords: Part 1 – uroscopy. *Clin Med.* 2001;1(6):507–9. <https://doi.org/10.7861/clinmedicine.1-6-507>
- 4 Folin O, Morris JL. On the determination of creatinine and creatine in urine. *J Biol Chem.* 1914;17(4):469–73. [https://doi.org/10.1016/s0021-9258\(18\)88386-7](https://doi.org/10.1016/s0021-9258(18)88386-7)
- 5 Kramer H, Rosas SE, Matsushita K. Beef tea, vitality, creatinine, and the estimated GFR. *Am J Kidney Dis.* 2016;67(2):169–72. <https://doi.org/10.1053/j.ajkd.2015.11.001>
- 6 Losito A. History of creatinine clearance: tribute to a forerunner. *Clin Kidney J.* 2023;16(6):891–5. <https://doi.org/10.1093/ckj/sfad024>
- 7 Delanghe JR, Speckaert MM. Creatinine determination according to Jaffe: what does it stand for? *NDT Plus.* 2011;4(2):83–6. <https://doi.org/10.1093/ndtplus/sfq211>
- 8 Hsu C, Chertow GM. Chronic renal confusion: insufficiency, failure, dysfunction, or disease. *Am J Kidney Dis.* 2000;36(2):415–8. <https://doi.org/10.1053/ajkd.2000.8996>
- 9 Strippoli GFM, Craig JC, Schena FP. The number, quality, and coverage of randomized controlled trials in nephrology. *J Am Soc Nephrol.* 2004;15(2):411–9. <https://doi.org/10.1097/01.asn.0000100125.21491.46>
- 10 Remuzzi G, Ruggenti P. Slowing the progression of diabetic nephropathy. *N Engl J Med.* 1993;329(20):1496–7. <https://doi.org/10.1056/NEJM19931113292012>
- 11 Levey AS, de Jong PE, Coresh J, El Nahas M, Astor BC, Matsushita K, et al. The definition, classification, and prognosis of chronic kidney disease: a KDIGO Controversies Conference report. *Kidney Int.* 2011;80(1):17–28. <https://doi.org/10.1038/ki.2010.483>
- 12 Levey AS, Coresh J, Balk E, Kausz AT, Levin A, Steffes MW, et al. National kidney foundation practice guidelines for chronic kidney disease: evaluation, classification, and stratification. *Ann Intern Med.* 2003;139(2):137–47. <https://doi.org/10.7326/0003-4819-139-2-200307150-00013>
- 13 Levin A, Stevens PE. Summary of KDIGO 2012 CKD Guideline: behind the scenes, need for guidance, and a framework for moving forward. *Kidney Int.* 2014;85(1):49–61. <https://doi.org/10.1038/ki.2013.444>
- 14 Coresh J, Selvin E, Stevens LA, Manzi J, Kusek JW, Eggers P, et al. Prevalence of chronic kidney disease in the United States. *JAMA.* 2007;298(17):2038–47. <https://doi.org/10.1001/jama.298.17.2038>
- 15 Mills KT, Xu Y, Zhang W, Bundy JD, Chen C-S, Kelly TN, et al. A systematic analysis of worldwide population-based data on the global burden of chronic kidney disease in 2010. *Kidney Int.* 2015;88(5):950–7. <https://doi.org/10.1038/ki.2015.230>
- 16 GBD Chronic Kidney Disease Collaboration, Purcell CA, Levey AS, Smith M, Abdoli A, Abebe M, et al. Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *Lancet.* 2020;395(10225):709–33. [https://doi.org/10.1016/S0140-6736\(20\)30045-3](https://doi.org/10.1016/S0140-6736(20)30045-3)
- 17 Perkovic V, Tuttle KR, Rossing P, Mahaffey KW, Mann JFE, Bakris G, et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med.* 2024;391:109–21. <https://doi.org/10.1056/NEJMoa2403347>
- 18 Bakris GL, Agarwal R, Anker SD, Pitt B, Ruilope LM, Rossing P, et al. Effect of finerenone on chronic kidney disease outcomes in type 2 diabetes. *N Engl J Med.* 2020;383(23):2219–29. <https://doi.org/10.1056/NEJMoa2025845>
- 19 Perkovic V, Jardine MJ, Neal B, Bompoint S, Heerspink HJL, Charytan DM, et al. Canagliflozin and renal outcomes in type 2 diabetes and nephropathy. *N Engl J Med.* 2019;380(24):2295–306. <https://doi.org/10.1056/NEJMoa1811744>
- 20 Abeysekera RA, Healy HG, Wang Z, Cameron AL, Hoy WE. Heterogeneity in patterns of progression of chronic kidney disease. *Intern Med J.* 2021;51(2):220–8. <https://doi.org/10.1111/imj.14770>
- 21 Ryu H, Hong Y, Kang E, Kang M, Kim J, Park HC, et al. Comparison of outcomes of chronic kidney disease based on etiology: a prospective cohort study from KNOW-CKD. *Sci Rep.* 2023;13(1):3570. <https://doi.org/10.1038/s41598-023-29844-x>
- 22 Palsson R, Short SAP, Kibbelaar ZA, Amodu A, Stillman IE, Rennke HG, et al. Bleeding complications after percutaneous native kidney biopsy: results from the Boston kidney biopsy cohort. *Kidney Int Rep.* 2020;5(4):511–8. <https://doi.org/10.1016/j.ekir.2020.01.012>
- 23 Hall YN, Himmelfarb J. The CKD classification system in the precision medicine era. *Clin J Am Soc Nephrol.* 2017;12(2):346–8. <https://doi.org/10.2215/CJN.09310916>
- 24 Kerr M, Bray B, Medcalf J, O'Donoghue DJ, Matthews B. Estimating the financial cost of chronic kidney disease to the NHS in England. *Nephrol Dial Transplant.* 2012;27(Suppl 3):iii73–80. <https://doi.org/10.1093/ndt/gfs269>
- 25 Canney M, Sood MM, Hundemer GL. Contemporary risk prediction models in chronic kidney disease: when less is more. *Curr Opin Nephrol Hypertens.* 2022;31(3):297–302. <https://doi.org/10.1097/MNH.0000000000000788>
- 26 Tangri N, Stevens LA, Griffith J, Tighiouart H, Djurdjev O, Naimark D, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305(15):1553–9. <https://doi.org/10.1001/jama.2011.451>

- 27 Tangri N, Grams ME, Levey AS, Coresh J, Appel LJ, Astor BC, et al. Multinational assessment of accuracy of equations for predicting risk of kidney failure: a meta-analysis. *JAMA*. 2016;315(2):164–74. <https://doi.org/10.1001/jama.2015.18202>
- 28 Dunkler D, Gao P, Lee SF, Heinze G, Clase CM, Tobe S, et al. Risk prediction for early CKD in type 2 diabetes. *Clin J Am Soc Nephrol*. 2015;10(8):1371–9. <https://doi.org/10.2215/CJN.10321014>
- 29 McDonnell T, Kalra PA, Vuilleumier N, Cockwell P, Wheeler DC, Fraser SDS, et al. The impact of primary renal diagnosis on prognosis and the varying predictive power of albuminuria in the NURTURE-CKD study. *Am J Nephrol*. 2024;1–12. <https://doi.org/10.1159/000541770>
- 30 Ali I, Donne RL, Kalra PA. A validation study of the kidney failure risk equation in advanced chronic kidney disease according to disease aetiology with evaluation of discrimination, calibration and clinical utility. *BMC Nephrol*. 2021;22(1):194. <https://doi.org/10.1186/s12882-021-02402-1>
- 31 Wong K, Pitcher D, Braddon F, Downward L, Steenkamp R, Annear N, et al. Effects of rare kidney diseases on kidney failure: a longitudinal analysis of the UK National Registry of Rare Kidney Diseases (RaDaR) cohort. *Lancet*. 2024;403(10433):1279–89. [https://doi.org/10.1016/S0140-6736\(23\)02843-X](https://doi.org/10.1016/S0140-6736(23)02843-X)
- 32 Dubin RF, Deo R, Ren Y, Wang J, Zheng Z, Shou H, et al. Proteomics of CKD progression in the chronic renal insufficiency cohort. *Nat Commun*. 2023;14(1):6340. <https://doi.org/10.1038/s41467-023-41642-7>
- 33 Al-Wahsh H, Tangri N, Quinn R, Liu P, Ferguson Ms T, Fiocco M, et al. Accounting for the competing risk of death to predict kidney failure in adults with stage 4 chronic kidney disease. *JAMA Netw Open*. 2021; 4(5):e219225. <https://doi.org/10.1001/jamanetworkopen.2021.9225>
- 34 Hill NR, Fatoba ST, Oke JL, Hirst JA, O'Callaghan CA, Lasserson DS, et al. Global prevalence of chronic kidney disease: a systematic review and meta-analysis. *PLoS One*. 2016;11(7):e0158765. <https://doi.org/10.1371/journal.pone.0158765>
- 35 Maher F, Teece L, Major RW, Bradbury N, Medcalf JF, Brunskill NJ, et al. Using the kidney failure risk equation to predict end-stage kidney disease in CKD patients of South Asian ethnicity: an external validation study. *Diagn Progn Res*. 2023;7(1):22. <https://doi.org/10.1186/s41512-023-00157-x>
- 36 Pitcher D, Braddon F, Hendry B, Mercer A, Osmaston K, Saleem MA, et al. Long-term outcomes in IgA nephropathy. *Clin J Am Soc Nephrol*. 2023;18(6):727–38. <https://doi.org/10.2215/CJN.0000000000000135>
- 37 Carrero JJ, Hecking M, Chesnaye NC, Jager KJ. Sex and gender disparities in the epidemiology and outcomes of chronic kidney disease. *Nat Rev Nephrol*. 2018;14(3):151–64. <https://doi.org/10.1038/nrneph.2017.181>
- 38 Hödlmoser S, Winkelmayr WC, Zee J, Pecoits-Filho R, Pisoni RL, Port FK, et al. Sex differences in chronic kidney disease awareness among US adults, 1999 to 2018. *PLoS One*. 2020;15(12):e0243431. <https://doi.org/10.1371/journal.pone.0243431>
- 39 Shabaka A, Cases-Corona C, Fernandez-Juarez G. Therapeutic insights in chronic kidney disease progression. *Front Med*. 2021;8:645187. <https://doi.org/10.3389/fmed.2021.645187>
- 40 Silver Spring (MD): Food and Drug Administration (US); Bethesda (MD): National Institutes of Health (US). BEST (biomarkers, EndpointS, and other tools) resource. 2016. [cited 2023 Oct 18]. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK326791/>
- 41 World Health organisation. Biomarkers in risk assessment: validity and validation. WHO International Programme on Chemical Safety; 2001.
- 42 Christensen EI, Birn H, Rippe B, Maunsbach AB. Controversies in nephrology: renal albumin handling, facts, and artifacts. *Kidney Int*. 2007;72(10):1192–4. <https://doi.org/10.1038/sj.ki.5002526>
- 43 Moeller MJ, Tenten V. Renal albumin filtration: alternative models to the standard physical barriers. *Nat Rev Nephrol*. 2013;9(5):266–77. <https://doi.org/10.1038/nrneph.2013.58>
- 44 Fogo AB. Mechanisms of progression of chronic kidney disease. *Pediatr Nephrol*. 2007;22(12):2011–22. <https://doi.org/10.1007/s00467-007-0524-0>
- 45 Levey AS, Gansevoort RT, Coresh J, Inker LA, Heerspink HL, Grams ME, et al. Change in albuminuria and GFR as end points for clinical trials in early stages of CKD: a scientific workshop sponsored by the national kidney foundation in collaboration with the US food and drug administration and European medicines agency. *Am J Kidney Dis*. 2020;75(1):84–104. <https://doi.org/10.1053/j.ajkd.2019.06.009>
- 46 Hannan M, Ansari S, Meza N, Anderson AH, Srivastava A, Waikar S, et al. Risk factors for CKD progression: overview of findings from the CRIC study. *Clin J Am Soc Nephrol*. 2021;16(4):648–59. <https://doi.org/10.2215/CJN.07830520>
- 47 Schrauben SJ, Shou H, Zhang X, Anderson AH, Bonventre JV, Chen J, et al. Association of multiple plasma biomarker concentrations with progression of prevalent diabetic kidney disease: findings from the chronic renal insufficiency cohort (CRIC) study. *J Am Soc Nephrol*. 2021;32(1):115–26. <https://doi.org/10.1681/ASN.2020040487>
- 48 Taal MW, Lucas B, Roderick P, Cockwell P, Wheeler DC, Saleem MA, et al. Associations with age and glomerular filtration rate in a referred population with chronic kidney disease: methods and baseline data from a UK multicentre cohort study (NURTURE-CKD). *Nephrol Dial Transplant*. 2023; 38(11):2617–26. <https://doi.org/10.1093/ndt/gfad110>
- 49 Yuan Q, Tang B, Zhang C. Signaling pathways of chronic kidney diseases, implications for therapeutics. *Signal Transduct Target Ther*. 2022;7(1):182. <https://doi.org/10.1038/s41392-022-01036-5>
- 50 Liu C, Debnath N, Mosoyan G, Chauhan K, Vasquez-Rios G, Soudant C, et al. Plasma and urine biomarkers for CKD outcomes: a systematic review and meta-analysis. *J Am Soc Nephrol*. 2022;33(11S):915. <https://doi.org/10.1681/asn.20223311s1915b>
- 51 Catanese L, Siwy J, Mischak H, Wendt R, Beige J, Rupprecht H. Recent advances in urinary peptide and proteomic biomarkers in chronic kidney disease: a systematic review. *Int J Mol Sci*. 2023;24(11):9156. <https://doi.org/10.3390/ijms24119156>
- 52 Barinotti A, Radin M, Cecchi I, Foddai SG, Rubini E, Roccatello D, et al. Serum biomarkers of renal fibrosis: a systematic review. *Int J Mol Sci*. 2022;23(22):14139. <https://doi.org/10.3390/ijms232214139>
- 53 Liu C, Debnath N, Mosoyan G, Chauhan K, Vasquez-Rios G, Soudant C, et al. Systematic review and meta-analysis of plasma and urine biomarkers for CKD outcomes. *J Am Soc Nephrol*. 2022;33(9):1657–72. <https://doi.org/10.1681/ASN.2022010098>
- 54 Martin WP, Tuohy C, Doody A, Jackson S, Canavan RJ, Slattery D, et al. Parallel assessment of albuminuria and plasma sTNFRI in people with type 2 diabetes and advanced chronic kidney disease provides accurate prognostication of the risks of renal decline and death. *Sci Rep*. 2020;10(1):14852. <https://doi.org/10.1038/s41598-020-71684-6>
- 55 Huber SA, Sakkinen P, Conze D, Hardin N, Tracy R. Interleukin-6 exacerbates early Atherosclerosis in mice. *Arterioscler Thromb Vasc Biol*. 1999;19(10):2364–7. <https://doi.org/10.1161/01.atv.19.10.2364>
- 56 Batra G, Ghukasyan Lakic T, Lindbäck J, Held C, White HD, Stewart RAH, et al. Interleukin 6 and cardiovascular outcomes in patients with chronic kidney disease and chronic coronary syndrome. *JAMA Cardiol*. 2021;6(12):1440–5. <https://doi.org/10.1001/jamacardio.2021.3079>
- 57 Liu Y, Xu K, Xiang Y, Ma B, Li H, Li Y, et al. Role of MCP-1 as an inflammatory biomarker in nephropathy. *Front Immunol*. 2023;14:1303076. <https://doi.org/10.3389/fimmu.2023.1303076>
- 58 Tuttle KR, Brosius FC, Adler SG, Kretzler M, Mehta RL, Tumlin JA, et al. JAK1/JAK2 inhibition by baricitinib in diabetic kidney disease: results from a Phase 2 randomized controlled clinical trial. *Nephrol Dial Transplant*. 2018;33(11):1950–9. <https://doi.org/10.1093/ndt/gfx377>
- 59 Ridker PM, MacFadyen JG, Glynn RJ, Koenig W, Libby P, Everett BM, et al. Inhibition of interleukin-1 β by canakinumab and cardiovascular outcomes in patients with Chronic Kidney disease. *J Am Coll Cardiol*. 2018;71(21):2405–14. <https://doi.org/10.1016/j.jacc.2018.03.490>

- 60 Schei J, Fuskevåg O-M, Stefansson VTN, Solbu MD, Jenssen TG, Eriksen BO, et al. Urinary markers of oxidative stress are associated with albuminuria but not GFR decline. *Kidney Int Rep.* 2018;3(3):573–82. <https://doi.org/10.1016/j.ekir.2017.11.020>
- 61 Roberts LJ, Morrow JD. Measurement of F2-isoprostanes as an index of oxidative stress in vivo. *Free Radic Biol Med.* 2000;28(4):505–13. [https://doi.org/10.1016/s0891-5849\(99\)00264-6](https://doi.org/10.1016/s0891-5849(99)00264-6)
- 62 Cottone S, Mule G, Guarneri M, Palermo A, Lorito MC, Riccobene R, et al. Endothelin-1 and F2-isoprostane relate to and predict renal dysfunction in hypertensive patients. *Nephrol Dial Transplant.* 2009;24(2):497–503. <https://doi.org/10.1093/ndt/gfn489>
- 63 Duni A, Liakopoulos V, Roumeliotis S, Peschos D, Dounousi E. Oxidative stress in the pathogenesis and evolution of chronic kidney disease: untangling ariadne's thread. *Int J Mol Sci.* 2019;20(15):3711. <https://doi.org/10.3390/ijms20153711>
- 64 Doria A, Galecki AT, Spino C, Pop-Busui R, Cherney DZ, Lingvay I, et al. Serum urate lowering with allopurinol and kidney function in type 1 diabetes. *N Engl J Med.* 2020;382(26):2493–503. <https://doi.org/10.1056/NEJMoa1916624>
- 65 Kimura K, Hosoya T, Uchida S, Inaba M, Makino H, Maruyama S, et al. Febuxostat therapy for patients with stage 3 CKD and asymptomatic hyperuricemia: a randomized trial. *Am J Kidney Dis.* 2018;72(6):798–810. <https://doi.org/10.1053/j.ajkd.2018.06.028>
- 66 Reutens AT, Jandeleit-Dahm K, Thomas M, Salim A, De Livera AM, Bach LA, et al. A physician-initiated double-blind, randomised, placebo-controlled, phase 2 study evaluating the efficacy and safety of inhibition of NADPH oxidase with the first-in-class Nox-1/4 inhibitor, GKT137831, in adults with type 1 diabetes and persistently elevated urinary albumin excretion: protocol and statistical considerations. *Contemp Clin Trials.* 2020;90:105892. <https://doi.org/10.1016/j.cct.2019.105892>
- 67 Frąk W, Kućmierz J, Szlagor M, Młynarska E, Rysz J, Franczyk B. New insights into molecular mechanisms of chronic kidney disease. *Biomedicines.* 2022;10(11):2846. <https://doi.org/10.3390/biomedicines10112846>
- 68 Cheng Z, Limbu M, Wang Z, Liu J, Liu L, Zhang X, et al. MMP-2 and 9 in chronic kidney disease. *Int J Mol Sci.* 2017;18(4):776. <https://doi.org/10.3390/ijms18040776>
- 69 Hu Q, Lan J, Liang W, Chen Y, Chen B, Liu Z, et al. MMP7 damages the integrity of the renal tubule epithelium by activating MMP2/9 during ischemia-reperfusion injury. *J Mol Histol.* 2020;51(6):685–700. <https://doi.org/10.1007/s10735-020-09914-4>
- 70 Zhou D, Tian Y, Sun L, Zhou L, Xiao L, Tan RJ, et al. Matrix metalloproteinase-7 is a urinary biomarker and pathogenic mediator of kidney fibrosis. *J Am Soc Nephrol.* 2017;28(2):598–611. <https://doi.org/10.1681/ASN.2016030354>
- 71 Wong MG, Perkovic V, Woodward M, Chalmers J, Li Q, Hillis GS, et al. Circulating bone morphogenetic protein-7 and transforming growth factor- β 1 are better predictors of renal end points in patients with type 2 diabetes mellitus. *Kidney Int.* 2013;83(2):278–84. <https://doi.org/10.1038/ki.2012.383>
- 72 Sharma K, Ix JH, Mathew AV, Cho M, Pflueger A, Dunn SR, et al. Pirfenidone for diabetic nephropathy. *J Am Soc Nephrol.* 2011;22(6):1144–51. <https://doi.org/10.1681/ASN.2010101049>
- 73 Vincenti F, Fervenza FC, Campbell KN, Diaz M, Gesualdo L, Nelson P, et al. A phase 2, double-blind, placebo-controlled, randomized study of fresolimumab in patients with steroid-resistant primary focal segmental glomerulosclerosis. *Kidney Int Rep.* 2017;2(5):800–10. <https://doi.org/10.1016/j.ekir.2017.03.011>
- 74 Song M-K, Lee J-H, Ryoo I, Lee S, Ku S-K, Kwak M-K. Bardoxolone ameliorates TGF- β 1-associated renal fibrosis through Nrf2/Smad7 elevation. *Free Radic Biol Med.* 2019;138:33–42. <https://doi.org/10.1016/j.freeradbiomed.2019.04.033>
- 75 de Zeeuw D, Akizawa T, Audhya P, Bakris GL, Chin M, Christ-Schmidt H, et al. Bardoxolone methyl in type 2 diabetes and stage 4 chronic kidney disease. *N Engl J Med.* 2013;369(26):2492–503. <https://doi.org/10.1056/NEJMoa1306033>
- 76 Panizo S, Martínez-Arias L, Alonso-Montes C, Cannata P, Martín-Carro B, Fernández-Martín JL, et al. Fibrosis in chronic kidney disease: pathogenesis and consequences. *Int J Mol Sci.* 2021;22(1):408. <https://doi.org/10.3390/ijms22010408>
- 77 Nezami BG, Farris AB. Assessing fibrosis in kidney biopsies. *Biomarkers in kidney disease.* Dordrecht, Netherlands: Springer; 2015; p. 1–21.
- 78 Wong MG, Pollock CA. Biomarkers in kidney fibrosis: are they useful? *Kidney Int Suppl.* (2011). 2014;4(1):79–83. <https://doi.org/10.1038/kisup.2014.15>
- 79 Zhang X, Chen X, Hong Q, Lin H, Zhu H, Liu Q, et al. TIMP-1 promotes age-related renal fibrosis through upregulating ICAM-1 in human TIMP-1 transgenic mice. *J Gerontol A Biol Sci Med Sci.* 2006;61(11):1130–43. <https://doi.org/10.1093/gerona/61.11.1130>
- 80 Zewinger S, Rauen T, Rudnicki M, Federico G, Wagner M, Triem S, et al. Dickkopf-3 (DKK3) in urine identifies patients with short-term risk of eGFR loss. *J Am Soc Nephrol.* 2018;29(11):2722–33. <https://doi.org/10.1681/ASN.2018040405>
- 81 Sánchez-Álamo B, García-Íñigo FJ, Shabaka A, Acedo JM, Cases-Corona C, Domínguez-Torres P, et al. Urinary Dickkopf-3: a new biomarker for CKD progression and mortality. *Nephrol Dial Transplant.* 2021;36(12):2199–207. <https://doi.org/10.1093/ndt/gfab198>
- 82 Zhou L, Li Y, Zhou D, Tan RJ, Liu Y. Loss of Klotho contributes to kidney injury by de-repression of Wnt/ β -catenin signaling. *J Am Soc Nephrol.* 2013;24(5):771–85. <https://doi.org/10.1681/ASN.2012080865>
- 83 Neyra JA, Hu MC. Potential application of klotho in human chronic kidney disease. *Bone.* 2017;100:41–9. <https://doi.org/10.1016/j.bone.2017.01.017>
- 84 Kanbay M, Brinza C, Ozbek L, Guldan M, Sisman U, Copur S, et al. The association between klotho and kidney and cardiovascular outcomes: a comprehensive systematic review and meta-analysis. *Clin Kidney J.* 2024;17(9):sfac255. <https://doi.org/10.1093/cckj/sfae255>
- 85 Schunk SJ, Floege J, Fliser D, Speer T. WNT- β -catenin signalling: a versatile player in kidney injury and repair. *Nat Rev Nephrol.* 2021;17(3):172–84. <https://doi.org/10.1038/s41581-020-00343-w>
- 86 Ix JH, Shlipak MG. The promise of tubule biomarkers in kidney disease: a review. *Am J Kidney Dis.* 2021;78(5):719–27. <https://doi.org/10.1053/j.ajkd.2021.03.026>
- 87 Griffin BR, You Z, Noureddine L, Gitomer B, Perrenoud L, Wang W, et al. KIM-1 and kidney disease progression in autosomal dominant polycystic kidney disease: HALT-PKD results. *Am J Nephrol.* 2020;51(6):473–9. <https://doi.org/10.1159/000508051>
- 88 Kuehn EW, Park KM, Somlo S, Bonventre JV. Kidney injury molecule-1 expression in murine polycystic kidney disease. *Am J Physiol Renal Physiol.* 2002;283(6):F1326–36. <https://doi.org/10.1152/ajprenal.00166.2002>
- 89 Romejko K, Markowska M, Niemczyk S. The review of current knowledge on neutrophil gelatinase-associated lipocalin (NGAL). *Int J Mol Sci.* 2023;24(13):10470. <https://doi.org/10.3390/ijms241310470>
- 90 Hsu C, Xie D, Waikar SS, Bonventre JV, Zhang X, Sabbiseti V, et al. Urine biomarkers of tubular injury do not improve on the clinical model predicting chronic kidney disease progression. *Kidney Int.* 2017;91(1):196–203. <https://doi.org/10.1016/j.kint.2016.09.003>
- 91 Greenberg JH, Abraham AG, Xu Y, Schelling JR, Feldman HI, Sabbiseti VS, et al. Plasma biomarkers of tubular injury and inflammation are associated with CKD progression in children. *J Am Soc Nephrol.* 2020;31(5):1067–77. <https://doi.org/10.1681/ASN.2019070723>
- 92 Weissinger EM, Wittke S, Kaiser T, Haller H, Bartel S, Krebs R, et al. Proteomic patterns established with capillary electrophoresis and mass spectrometry for diagnostic purposes. *Kidney Int.* 2004;65(6):2426–34. <https://doi.org/10.1111/j.1523-1755.2004.00659.x>

- 93 Good DM, Zürbig P, Argilés A, Bauer HW, Behrens G, Coon JJ, et al. Naturally occurring human urinary peptides for use in diagnosis of chronic kidney disease. *Mol Cell Proteomics*. 2010;9(11):2424–37. <https://doi.org/10.1074/mcp.M110.001917>
- 94 Tofte N, Lindhardt M, Adamova K, Bakker SJL, Beige J, Beulens JWJ, et al. Early detection of diabetic kidney disease by urinary proteomics and subsequent intervention with spironolactone to delay progression (PRIORITY): a prospective observational study and embedded randomised placebo-controlled trial. *Lancet Diabetes Endocrinol*. 2020;8(4):301–12. [https://doi.org/10.1016/S2213-8587\(20\)30026-7](https://doi.org/10.1016/S2213-8587(20)30026-7)
- 95 Lindhardt M, Persson F, Zürbig P, Stalmach A, Mischak H, de Zeeuw D, et al. Urinary proteomics predict onset of microalbuminuria in normoalbuminuric type 2 diabetic patients, a sub-study of the DIRECT-Protect 2 study. *Nephrol Dial Transplant*. 2016;gfw292. <https://doi.org/10.1093/ndt/gfw292>
- 96 Schanstra JP, Zürbig P, Alkhalaf A, Argiles A, Bakker SJL, Beige J, et al. Diagnosis and prediction of CKD progression by assessment of urinary peptides. *J Am Soc Nephrol*. 2015;26(8):1999–2010. <https://doi.org/10.1681/ASN.2014050423>
- 97 Pontillo C, Jacobs L, Staessen JA, Schanstra JP, Rossing P, Heerspink HJL, et al. A urinary proteome-based classifier for the early detection of decline in glomerular filtration. *Nephrol Dial Transplant*. 2016;gfw239. <https://doi.org/10.1093/ndt/gfw239>
- 98 Ålund O, Unwin R, Challis B, Kalra PA, Taal MW, Wheeler DC, et al. A note on performance metrics for the kidney failure risk equation. *Nephrol Dial Transplant*. 2023;39(9):1523–5. <https://doi.org/10.1093/ndt/gfae098>
- 99 Takeuchi K, Naito S, Kawashima N, Ishigaki N, Sano T, Kamata K, et al. New anti-nephrin antibody mediated podocyte injury model using a C57bl/6 mouse strain. *Nephron*. 2018;138(1):71–87. <https://doi.org/10.1159/000479935>
- 100 Watts AJB, Keller KH, Lerner G, Rosales I, Collins AB, Sekulic M, et al. Discovery of autoantibodies targeting nephrin in minimal change disease supports a novel autoimmune etiology. *J Am Soc Nephrol*. 2022;33(1):238–52. <https://doi.org/10.1681/ASN.2021060794>
- 101 Beck LH, Bonegio RGB, Lambeau G, Beck DM, Powell DW, Cummins TD, et al. M-Type phospholipase A2 receptor as target antigen in idiopathic membranous nephropathy. *N Engl J Med*. 2009;361(1):11–21. <https://doi.org/10.1056/NEJMoa0810457>
- 102 Pontillo C, Mischak H. Urinary biomarkers to predict CKD: is the future in multi-marker panels? *Nephrol Dial Transplant*. 2016;31(9):1373–5. <https://doi.org/10.1093/ndt/gfv467>
- 103 Krzyszczyk P, Acevedo A, Davidoff EJ, Timmins LM, Marrero-Berrios I, Patel M, et al. The growing role of precision and personalized medicine for cancer treatment. *Technol (Singap World Sci)*. 2018;6(3–4):79–100. <https://doi.org/10.1142/S2339547818300020>
- 104 Speer T, Schunk SJ, Sarakpi T, Schmit D, Wagner M, Arnold L, et al. Urinary DKK3 as a biomarker for short-term kidney function decline in children with chronic kidney disease: an observational cohort study. *Lancet Child Adolesc Health*. 2023;7(6):405–14. [https://doi.org/10.1016/S2352-4642\(23\)00049-4](https://doi.org/10.1016/S2352-4642(23)00049-4)
- 105 Andersen S, Mischak H, Zürbig P, Parving H-H, Rossing P. Urinary proteome analysis enables assessment of renoprotective treatment in type 2 diabetic patients with microalbuminuria. *BMC Nephrol*. 2010;11(1):29. <https://doi.org/10.1186/1471-2369-11-29>
- 106 Lindhardt M, Persson F, Oxlund C, Jacobsen IA, Zürbig P, Mischak H, et al. Predicting albuminuria response to spironolactone treatment with urinary proteomics in patients with type 2 diabetes and hypertension. *Nephrol Dial Transplant*. 2017;gfw406. <https://doi.org/10.1093/ndt/gfw406>
- 107 Biomarkers Definitions Working Group. Biomarkers and surrogate endpoints: preferred definitions and conceptual framework. *Clin Pharmacol Ther*. 2001;69(3):89–95. <https://doi.org/10.1067/mcp.2001.113989>
- 108 Levey AS, Catran D, Friedman A, Miller WG, Sedor J, Tuttle K, et al. Proteinuria as a surrogate outcome in CKD: report of a scientific workshop sponsored by the national kidney foundation and the US food and drug administration. *Am J Kidney Dis*. 2009;54(2):205–26. <https://doi.org/10.1053/j.ajkd.2009.04.029>
- 109 Heerspink HJL, Radhakrishnan J, Alpers CE, Barratt J, Bieler S, Diva U, et al. Spar-sentan in patients with IgA nephropathy: a prespecified interim analysis from a randomised, double-blind, active-controlled clinical trial. *Lancet*. 2023;401(10388):1584–94. [https://doi.org/10.1016/S0140-6736\(23\)00569-X](https://doi.org/10.1016/S0140-6736(23)00569-X)
- 110 Lafayette R, Kristensen J, Stone A, Floege J, Tesar V, Trimarchi H, et al. Efficacy and safety of a targeted-release formulation of budesonide in patients with primary IgA nephropathy (NefIgArd): 2-year results from a randomised phase 3 trial. *Lancet*. 2023;402(10405):859–70. [https://doi.org/10.1016/S0140-6736\(23\)01554-4](https://doi.org/10.1016/S0140-6736(23)01554-4)
- 111 Bagchi S, Lingaiah R, Mani K, Barwad A, Singh G, Balooni V, et al. Significance of serum galactose deficient IgA1 as a potential biomarker for IgA nephropathy: a case control study. *PLoS One*. 2019;14(3):e0214256. <https://doi.org/10.1371/journal.pone.0214256>
- 112 Molyneux K, Nawaz N, Taylor S, Barratt J. Treatment with nefecon reduces circulating levels of galactose-deficient IgA1 in patients with IgA nephropathy in the NefIgArd clinical trial. *J Am Soc Nephrol*. 2022;33(11S):783–4. <https://doi.org/10.1681/asn.20223311s1783d>
- 113 Cheung CK, Barratt J, Liew A, Zhang H, Tesar V, Lafayette R. The role of BAFF and APRIL in IgA nephropathy: pathogenic mechanisms and targeted therapies. *Front Nephrol*. 2023;3:1346769. <https://doi.org/10.3389/fneph.2023.1346769>
- 114 Tortajada A, Gutierrez E, Pickering MC, Praga Terente M, Medjeral-Thomas N. The role of complement in IgA nephropathy. *Mol Immunol*. 2019;114:123–32. <https://doi.org/10.1016/j.molimm.2019.07.017>
- 115 Zhuang Y, Lu H, Li J. Advances in the treatment of IgA nephropathy with biological agents. *Chronic Dis Transl Med*. 2024;10(1):1–11. <https://doi.org/10.1002/cdt3.104>
- 116 Selvaskandan H, Shi S, Twaij S, Cheung CK, Barratt J. Monitoring immune responses in IgA nephropathy: biomarkers to guide management. *Front Immunol*. 2020;11:572754. <https://doi.org/10.3389/fimmu.2020.572754>
- 117 Kerr KF, Roth J, Zhu K, Thiessen-Philbrook H, Meisner A, Wilson FP, et al. Evaluating biomarkers for prognostic enrichment of clinical trials. *Clin Trials*. 2017;14(6):629–38. <https://doi.org/10.1177/1740774517723588>
- 118 Heerspink HJL, Parving H-H, Andress DL, Bakris G, Correa-Rotter R, Hou F-F, et al. Atrasentan and renal events in patients with type 2 diabetes and chronic kidney disease (SONAR): a double-blind, randomised, placebo-controlled trial. *Lancet*. 2019;393(10184):1937–47. [https://doi.org/10.1016/S0140-6736\(19\)30772-X](https://doi.org/10.1016/S0140-6736(19)30772-X)
- 119 Lafayette RA, Rovin BH, Reich HN, Tumlin JA, Floege J, Barratt J. Safety, tolerability and efficacy of narsoplimab, a novel MASP-2 inhibitor for the treatment of IgA nephropathy. *Kidney Int Rep*. 2020;5(11):2032–41. <https://doi.org/10.1016/j.ekir.2020.08.003>
- 120 Rizk DV, Rovin BH, Zhang H, Kashihara N, Maes B, Trimarchi H, et al. Targeting the alternative complement pathway with iptacopan to treat IgA nephropathy: design and rationale of the APPLAUSE-IgAN study. *Kidney Int Rep*. 2023;8(5):968–79. <https://doi.org/10.1016/j.ekir.2023.01.041>
- 121 Neuen BL, Kennedy S. Biomarkers and personalised medicine in paediatric kidney disease. *Lancet Child Adolesc Health*. 2023;7(6):369–71. [https://doi.org/10.1016/S2352-4642\(23\)00102-5](https://doi.org/10.1016/S2352-4642(23)00102-5)
- 122 Varo N, Nuzzo R, Natal C, Libby P, Schönbeck U. Influence of pre-analytical and analytical factors on soluble CD40L measurements. *Clin Sci*. 2006;111(5):341–7. <https://doi.org/10.1042/CS20060047>

- 123 Wenzel F, Baertl A, Zimmermann N, Hohlfeld T, Giers G, Oldenburg J, et al. Different behaviour of soluble CD40L concentrations can be reflected by variations of preanalytical conditions. *Clin Hemorheol Microcirc.* 2008;39(1-4):417–22. <https://doi.org/10.3233/ch-2008-1112>
- 124 Bouwmeester W, Zuithoff NPA, Mallett S, Geerlings MI, Vergouwe Y, Steyerberg EW, et al. Reporting and methods in clinical prediction research: a systematic review. *PLoS Med.* 2012;9(5):e1001221–12. <https://doi.org/10.1371/journal.pmed.1001221>
- 125 Mallett S, Royston P, Dutton S, Waters R, Altman DG. Reporting methods in studies developing prognostic models in cancer: a review. *BMC Med.* 2010;8:20. <https://doi.org/10.1186/1741-7015-8-20>
- 126 White N, Parsons R, Collins G, Barnett A. Evidence of questionable research practices in clinical prediction models. *BMC Med.* 2023;21(1):339. <https://doi.org/10.1186/s12916-023-03048-6>
- 127 Mucherino S, Lorenzoni V, Orlando V, Triulzi I, Del Re M, Capuano A, et al. Cost-effectiveness of treatment optimisation with biomarkers for immunotherapy in solid tumours: a systematic review protocol. *BMJ Open.* 2021;11(9):e048141. <https://doi.org/10.1136/bmjopen-2020-048141>
- 128 Stanzick KJ, Li Y, Schlosser P, Gorski M, Wuttke M, Thomas LF, et al. Discovery and prioritization of variants and genes for kidney function in >1.2 million individuals. *Nat Commun.* 2021;12(1):4350. <https://doi.org/10.1038/s41467-021-24491-0>
- 129 Lanktree MB, Perrot N, Smyth A, Chong M, Narula S, Shanmuganathan M, et al. A novel multi-ancestry proteome-wide Mendelian randomization study implicates extracellular proteins, tubular cells, and fibroblasts in estimated glomerular filtration rate regulation. *Kidney Int.* 2023;104(6):1170–84. <https://doi.org/10.1016/j.kint.2023.08.025>
- 130 Yao C, Chen G, Song C, Keefe J, Mendelson M, Huan T, et al. Genome-wide mapping of plasma protein QTLs identifies putatively causal genes and pathways for cardiovascular disease. *Nat Commun.* 2018;9(1):3268. <https://doi.org/10.1038/s41467-018-05512-x>
- 131 Schlosser P, Grams ME, Rhee EP. Proteomics: progress and promise of high-throughput proteomics in chronic kidney disease. *Mol Cell Proteomics.* 2023;22(6):100550. <https://doi.org/10.1016/j.mcpro.2023.100550>
- 132 Ferkingstad E, Sulem P, Atlason BA, Sveinbjornsson G, Magnusson MI, Styrismisdottir EL, et al. Large-scale integration of the plasma proteome with genetics and disease. *Nat Genet.* 2021;53(12):1712–21. <https://doi.org/10.1038/s41588-021-00978-w>
- 133 Coca SG. “Scanning” into the future: the promise of SOMAScan technology for kidney disease. *Kidney Int Rep.* 2018;3(5):1020–2. <https://doi.org/10.1016/j.ekir.2018.06.009>
- 134 Niewczas MA, Pavkov ME, Skupien J, Smiles A, Md Dom ZI, Wilson JM, et al. A signature of circulating inflammatory proteins and development of end-stage renal disease in diabetes. *Nat Med.* 2019;25(5):805–13. <https://doi.org/10.1038/s41591-019-0415-5>
- 135 Fountzilas E, Tsimberidou AM, Vo HH, Kurzrock R. Clinical trial design in the era of precision medicine. *Genome Med.* 2022;14(1):101. <https://doi.org/10.1186/s13073-022-01102-1>
- 136 Steuer C, Papadimitrakopoulou V, Herbst R, Redman M, Hirsch F, Mack P, et al. Innovative clinical trials: the LUNG-MAP study. *Clin Pharmacol Ther.* 2015;97(5):488–91. <https://doi.org/10.1002/cpt.88>
- 137 O'Dwyer PJ, Gray RJ, Flaherty KT, Chen AP, Li S, Wang V, et al. The NCI-MATCH trial: lessons for precision oncology. *Nat Med.* 2023;29(6):1349–57. <https://doi.org/10.1038/s41591-023-02379-4>